Active Rail Reduced-Model Translation to ADM

Technical Write-Up and Prototype Plan

Draft prepared for Daniel Goldman

May 16, 2026

Abstract

This draft records the present engineering status of the active-rail reduced model and specifies a prototype path for an ADM/3+1 scaffold. The operating target is set to $V = 5$ for primary development. The $V = 10$ branch is retained as the recognized edge case for the current design. The main result is that the exact field-builder ADM pass survives at $V = 5$: live packet norm remains clean, live negative ADM energy density is absent, and the apparent packet-local $|\rho_{\mathrm{ADM}}|$ burden subtracts away as standing substrate curvature when compared against a time-matched beta-off substrate. The remaining live service increment is primarily a catch/rematch radial momentum handoff, represented in ADM by Δj_l, and it scales upward at $V = 10$ without producing a new packet-filling ρ failure. This is a reduced, prescribed-geometry result. It is not a physical matter construction, a semiclassical stress-tensor calculation, or a solved initial-data set.

1 Engineering status

The active rail is treated here as a source-placement architecture: a prepared standing support geometry carries the baseline throat/angular substrate, while the live service cycle injects a controlled shift handoff through entry, catch/rematch, carried shift, fade, and reset. The current frozen branch is

Design label	R1.75 shaped catch radial-soft lapse-cushion v1
Primary target	$V = 5$
Recognized edge	$V = 10$
Passenger radius	$R_{\mathrm{pass}} = 0.35$
Support/quarantine radius	$R_{\mathrm{th}} = 1.75$
Angular jacket radius	$R_{\Omega} = 1.75$
Radial support width	$w_{\mathrm{th}} = 0.569$
Packet transition width	$w_{\mathrm{pass}} = 0.06$
Angular jacket width/amplitude	$w_{\Omega} = 1.4$, $a_{\Omega} = 0.20$
Lapse cushion	$\eta_N = 2.00$
Catch/rematch	minimum-jerk locked-lead, support $x = -0.05$, packet $x = 0.00$, width 0.32

The prior reduced-source validation established three practical gates. First, grid checks on the locked branch did not reveal a packet norm failure at the design point. Second, targeted off-axis/null screening did not reveal a hidden angled-ray packet-edge catastrophe. Third, residual maps localized the remaining live radial-null underfit near the inner packet edge during catch/rematch. These gates justify an ADM scaffold, not a physical-source claim.

2 Metric form and ADM decomposition

The reduced rail metric is written as

$$\mathrm{d}s^2 = -\alpha(l,\sigma)^2\mathrm{d}\sigma^2 + A(l,\sigma)\left(\mathrm{d}l + \beta(l,\sigma)\mathrm{d}\sigma\right)^2 + B(l,\sigma)\mathrm{d}\Omega^2, \tag{1}$$

where $A = \gamma_{ll}$, $B = \gamma_{\Omega\Omega}$, and $\beta = \beta^l$. This is already in 3+1 form,

$$\mathrm{d}s^2 = -\alpha^2\mathrm{d}\sigma^2 + \gamma_{ij}(\mathrm{d}x^i + \beta^i\mathrm{d}\sigma)(\mathrm{d}x^j + \beta^j\mathrm{d}\sigma), \tag{2}$$

with

$$\gamma_{ij} = \mathrm{diag}\left(A, B, B\sin^2\theta\right), \qquad \beta^i = (\beta, 0, 0), \qquad \beta_i = (A\beta, 0, 0). \tag{3}$$

The normal observer is

$$n^\mu = \left(\frac{1}{\alpha}, -\frac{\beta}{\alpha}, 0, 0\right), \qquad n_\mu = (-\alpha, 0, 0, 0). \tag{4}$$

The extrinsic curvature convention used in this draft is

$$K_{ij} = \frac{1}{2\alpha}\left(D_i\beta_j + D_j\beta_i - \partial_\sigma\gamma_{ij}\right). \tag{5}$$

Primes denote ∂_l and overdots denote ∂_σ. The nonzero independent components are

$$K_{ll} = \frac{1}{2\alpha}\left(2A\beta' + A'\beta - \dot{A}\right), \tag{6}$$

$$K_{\theta\theta} = \frac{1}{2\alpha}\left(\beta B' - \dot{B}\right), \tag{7}$$

$$K_{\phi\phi} = K_{\theta\theta}\sin^2\theta. \tag{8}$$

It is convenient to use the mixed eigenvalues

$$k_l = K^l{}_l = \frac{1}{2\alpha}\left(2\beta' + \beta\frac{A'}{A} - \frac{\dot{A}}{A}\right), \tag{9}$$

$$k_\Omega = K^\theta{}_\theta = K^\phi{}_\phi = \frac{1}{2\alpha}\left(\beta\frac{B'}{B} - \frac{\dot{B}}{B}\right), \tag{10}$$

so that

$$K = k_l + 2k_\Omega, \qquad K_{ij}K^{ij} = k_l^2 + 2k_\Omega^2. \tag{11}$$

For the spatial line element $\mathrm{d}l_3^2 = A\mathrm{d}l^2 + B\mathrm{d}\Omega^2$, the scalar curvature used in the scaffold is

$${}^{(3)}R = -\frac{2B''}{AB} + \frac{(B')^2}{2AB^2} + \frac{A'B'}{A^2B} + \frac{2}{B}. \tag{12}$$

The ADM constraint ledger is then

$$16\pi\rho_{\mathrm{ADM}} = {}^{(3)}R + K^2 - K_{ij}K^{ij} = {}^{(3)}R + 4k_lk_\Omega + 2k_\Omega^2, \tag{13}$$

$$8\pi j_l = D_j\left(K^j{}_l - \delta^j_l K\right) = -2k'_\Omega + \frac{B'}{B}(k_l - k_\Omega). \tag{14}$$

Equations (13) and (14) are the first computational gate. Pressure channels and full null projections require additional projections of the full demanded Einstein tensor; they are deferred until the constraint scaffold is stable.

3 Validation evidence feeding the ADM scaffold

3.1 Locked model convergence at the edge

The locked V=10 reduced-source grid checks remain packet-safe across the tested grids. The relevant edge-case numbers are summarized below.

Grid	Max live packet norm	Positive packet norm points	Live $\lvert p_l \rvert$	Live negative radial T_{kk}
25x45	-198.925	0	13.8019	71.4351
33x59	-198.925	0	13.9880	72.5353
41x73	-198.925	0	13.8318	72.3794

This is a finite-difference stability check, not a Richardson convergence proof. Its engineering use is narrower: it says the frozen edge case does not fail immediately under ordinary grid refinement in the reduced ledger.

3.2 Source-risk scaling with velocity target

At $V = 5$, the source screen is materially less severe than at $V = 10$. The radial pressure burden is essentially fixed by the standing support geometry, while live negative radial T_{kk} falls substantially.

Target	Source basis	Max live packet norm	Live $\lvert p_l \rvert$	Live negative radial T_{kk}
$V = 5$	archived V-sweep	-251.331	13.8318	52.0568
$V = 9.5$	interpolated	-204.166	13.8318	70.3471
$V = 10$	archived V-sweep	-198.925	13.8318	72.3794

This motivates the engineering decision used in this draft: $V = 5$ is the primary prototype target, and $V = 10$ is the stress-test edge.

3.3 Residual localization and absorber ansatz

The residual screen identifies the live radial-null catch/rematch underfit as the main remaining source-closure issue. The residual is localized near the inner packet boundary, approximately $\xi = l - s(\sigma) \approx -R_{\text{pass}}$, and not smeared through the entire passenger packet. A compact source-control law should therefore be packet-adjacent and catch-gated rather than a global exotic layer. The tested ansatz class is

$$A_{\text{catch}}(\xi, \sigma) = W_{\text{catch}}(\sigma) \left[a H_{\text{shift}}(\xi, \sigma) + b B_{\text{inner}}(\xi) \right], \tag{15}$$

where W_{catch} is the locked-lead catch/rematch timing gate, H_{shift} is a smooth shift-handoff or mismatch measure, and B_{inner} is a narrow envelope centered on the inward packet edge.

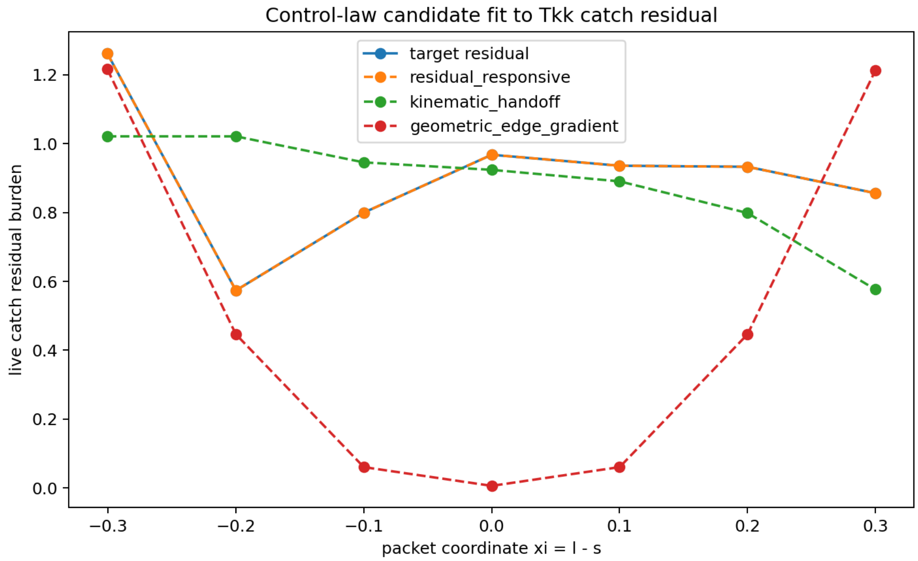


Figure 1: Proxy control-law comparison against the extracted live catch/rematch residual bins. The kinematic handoff law is the preferred primary ansatz; the residual-responsive law is useful as a diagnostic limiter but too self-referential as a physical law.

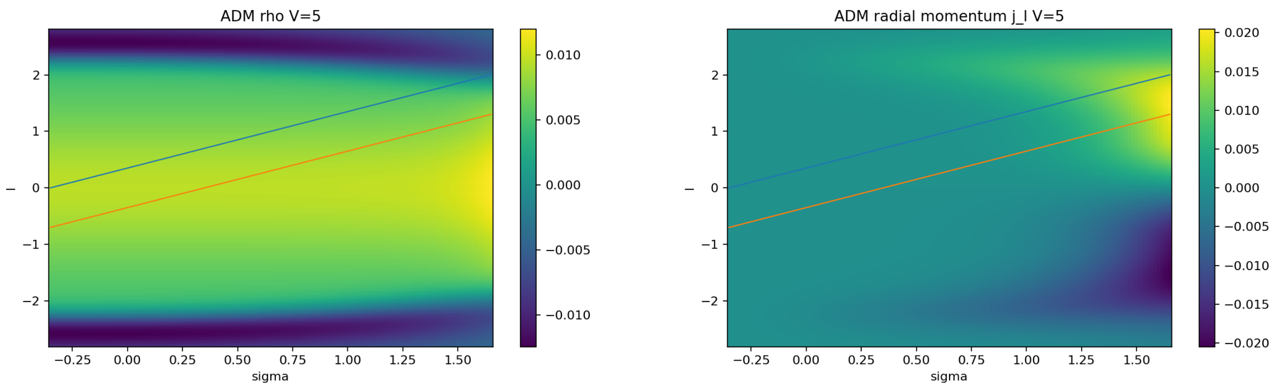
Candidate	Target coverage	Remaining underfit	Oversupply	Use
Residual-responsive	100.0%	0.00%	0.00%	cap/limiter
Kinematic handoff	88.2%	1.17%	0.93%	primary ansatz
Geometric edge-gradient	48.9%	5.07%	0.56%	envelope only

4 Exact-builder ADM survival check

The reconstructed ADM scaffold was replaced by an exact-builder pass using the original field definitions from `radial_softening_probe.py`. That exact-builder pass evaluates the same α, β, A, B, packet masks, stage masks, and region masks used by the final reduced-field construction. This closes the main reproducibility caveat of the initial ADM scaffold, while still leaving the broader physical-source caveat in place.

V	Channel	Scope	Burden	Fraction of global
5	$\lvert\rho_{\mathrm{ADM}}\rvert$	live packet total	13.5106	27.06%
5	negative ρ_{ADM}	live packet total	0.0000	0.00%
5	$\lvert j_l\rvert$	live packet total	0.1713	7.85%
5	$\lvert j_l\rvert$	live support shell	1.5084	69.14%
10	$\lvert\rho_{\mathrm{ADM}}\rvert$	live packet total	13.5106	27.06%
10	negative ρ_{ADM}	live packet total	0.0000	0.00%

The live packet $|\rho_{\mathrm{ADM}}|$ fraction initially looks high. The absence of live negative ρ_{ADM} and the weak dependence on V indicate that this is mostly standing throat/angular curvature. The substrate subtraction in the next section confirms that reading.



(a) ρ_{ADM}, exact-builder $V = 5$.

(b) j_l, exact-builder $V = 5$.

Figure 2: First exact-builder ADM maps at the primary target. The key interpretation is not the absolute ρ_{ADM} map alone, but the substrate-subtracted increment.

5 Standing substrate subtraction

The standing-substrate subtraction compares the live service geometry against a time-matched beta-off substrate. Operationally, the subtraction defines

$$\Delta\rho_{\mathrm{ADM}} = \rho_{\mathrm{live}}(\alpha, \beta, A, B) - \rho_{\mathrm{sub}}(\alpha, 0, A, B), \tag{16}$$

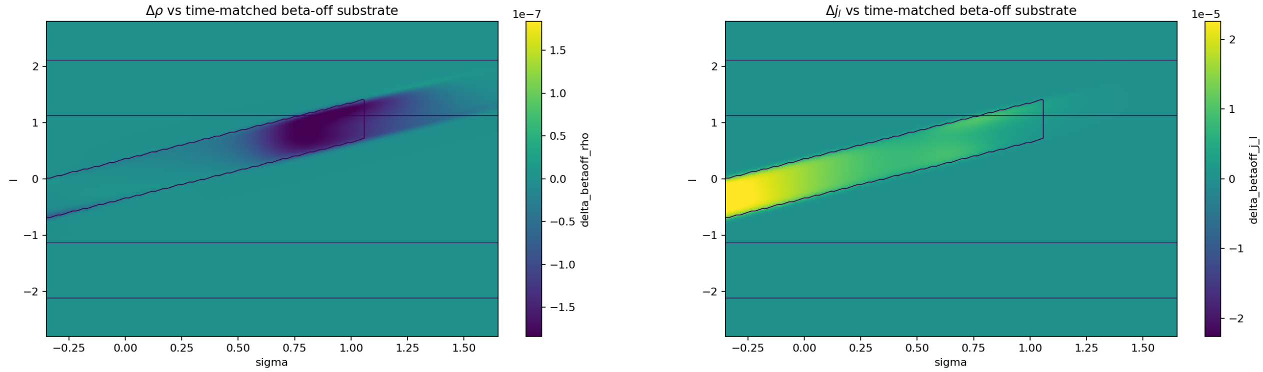
$$\Delta j_l = j_{l,\mathrm{live}}(\alpha, \beta, A, B) - j_{l,\mathrm{sub}}(\alpha, 0, A, B). \tag{17}$$

This is a diagnostic subtraction. It does not solve the standing substrate; it separates the shift/service-induced increment from the prepared support curvature inside the prescribed geometry.

At $V = 5$, the apparent packet $|\rho_{\mathrm{ADM}}|$ concern nearly vanishes under this subtraction. The time-matched beta-off $\Delta\rho_{\mathrm{ADM}}$ is 9.28×10^{-7} of the live global $|\rho_{\mathrm{ADM}}|$ burden and 3×10^{-6} of the original live packet $|\rho_{\mathrm{ADM}}|$ burden. The live service signal is Δj_l, not $\Delta\rho$.

Target	Field	Global $\lvert\Delta\rvert$	Live packet $\lvert\Delta\rvert$	Fraction of live global	Fraction of live packet live
$V = 5$	$\Delta\rho_{\mathrm{ADM}}$	4.6×10^{-5}	4.0×10^{-5}	9.28×10^{-7}	3.0×10^{-6}
$V = 5$	Δj_l	0.02145	0.02013	0.00983	0.11753
$V = 10$	$\Delta\rho_{\mathrm{ADM}}$	6.8×10^{-5}	5.6×10^{-5}	1.35×10^{-6}	4.0×10^{-6}
$V = 10$	Δj_l	0.03829	0.03597	0.01742	0.19250

The $V = 10$ edge case preserves the same qualitative pattern. $\Delta\rho_{\mathrm{ADM}}$ remains tiny. Δj_l increases by about $1.79\times$ relative to $V = 5$, and about 97.9% of the live-packet Δj_l increment lies in catch/rematch. The edge case is therefore sharper in the expected service channel rather than producing a new energy-density failure.



(a) $\Delta\rho_{\rm ADM}$ against beta-off substrate, $V = 5$.

(b) Δj_l against beta-off substrate, $V = 5$.

Figure 3: Substrate-subtracted service increments at the primary target. The energy-density increment is tiny; the service increment is the radial momentum handoff.

6 Physical plausibility screen

The source plausibility screen remains mixed. The favorable engineering result is localization: the live service increment is not a broad passenger-packet energy-density demand. It is primarily a packet-adjacent catch/rematch momentum channel. That channel is still a serious physics gate because compact null/momentum support near a packet edge is where quantum-inequality and thin-wall constraints become severe in related warp and wormhole contexts.

The literature baseline is appropriately conservative. Alcubierre's original warp-drive construction shows that a prescribed metric can demand exotic matter even when local travel inside the bubble is ordinary in the usual sense [1]. Pfenning and Ford applied quantum-inequality restrictions to the Alcubierre metric and found bubble-wall thickness requirements on the order of a few hundred Planck lengths, with physically unattainable integrated negative energy in that setup [2]. Barcelo and Visser showed that nonminimally coupled scalar fields can violate energy conditions and support wormhole branches, while also finding trans-Planckian scalar field values in those wormhole solutions [3]. Bobrick and Martire emphasize a broader classification in which warp-drive spacetimes are material shells, with subluminal positive-energy cases possible and superluminal cases still demanding careful energy-condition analysis [4].

For this project, the immediate claim is narrower:

> The active-rail reduced model localizes the live service increment primarily into a packet-adjacent catch/rematch radial momentum channel, while the apparent live packet ADM energy-density burden is mostly standing substrate curvature under beta-off subtraction.

This is an engineering target for a source family, not a completed physical source construction.

7 Prototype plan

7.1 Prototype A: ADM diagnostic hardening at $V = 5$

1. Freeze the exact field-builder import as a named module: `active_rail_fields.py`.
2. Recompute α, β, A, B, stage masks, packet masks, and support masks from the original builder.
3. Recompute K_{ij}, K, ${}^{(3)}R$, $\rho_{\rm ADM}$, and j_l using equations (13) and (14).

4. Run grid refinement at $V = 5$ first. Use $V = 10$ only as an edge appendix.
5. Produce substrate-subtracted maps for $\Delta\rho_{\mathrm{ADM}}$ and Δj_l.

Exit criterion: $\Delta\rho_{\mathrm{ADM}}$ stays small in the packet, Δj_l remains catch/rematch localized, and no packet norm failure appears under refinement.

7.2 Prototype B: service-induced source isolation

1. Define three substrates: static-ready, beta-off time-matched, and no-catch/no-rematch hold.
2. Decompose increments into standing substrate, shift-handoff, catch/rematch, release/fade, and reset.
3. Build a normalized service ledger with fractions by region and stage.
4. Treat live packet interior, packet edge, support shell, and outer jacket as separate engineering zones.

Exit criterion: the service-induced increment is dominated by support shell and packet-adjacent catch/rematch zones rather than passenger-packet interior.

7.3 Prototype C: compact catch/rematch source law

Implement equation (15) with a kinematic handoff amplitude and inner-edge envelope. The first model should be nonnegative and capped:

$$0 \leq A_{\mathrm{catch}}(\xi, \sigma) \leq C \,\mathrm{smooth}\left([R_{Tkk}(\xi, \sigma)]_+\right), \tag{18}$$

where the cap is a diagnostic limiter rather than the physical law. The physical ansatz should be driven by H_{shift} and shaped by B_{inner}.

Exit criterion: the compact law covers most of the catch/rematch residual without becoming a broad global layer, without shifting the residual into the packet interior, and without increasing packet norm risk.

7.4 Prototype D: first constraint-compatible source-family search

The first source-family search should be constrained by the ADM ledger rather than the earlier reduced proxy alone. Candidate families:

- standing support shell for baseline throat/angular curvature;
- effective angular jacket for the low-live-exposure angular burden;
- scalar-tensor or nonminimally coupled scalar comparison family for energy-condition-violating channels;
- controlled effective stress component for the catch/rematch momentum channel.

Exit criterion: at least one source decomposition covers the $V = 5$ ADM increments without trans-Planckian local amplitudes or resolution-growing packet-edge spikes.

8 Risk register

Risk item	Status	Engineering read	Gate
Standing substrate physical source	open	The subtraction shows it is baseline support curvature, but it still needs a source model or solved initial data.	high
Catch/rematch Δj_l	active	Correctly localized and scalable; the principal live service channel.	high
Compact null-stress absorber	active	Kinematic law is plausible as a control ansatz; physical realization is unproven.	high
Packet $\Delta\rho$	favorable	Tiny after beta-off subtraction at $V = 5$ and $V = 10$.	low
V=10 headroom	warning	Edge case works at the target point but has little margin above it.	medium
Angular/current/density proxy underfit	open	Mostly infrastructure-side or live-small in the current ledger.	medium
Exact reproducibility	improving	Exact field builder was imported for ADM maps; next step is clean module packaging.	medium

9 Recommended next deliverables

The next deliverables should be separated cleanly from exploratory bundles:

1. **ADM scaffold repository module**: exact field builder, ADM derivative operators, constraints, masks, and plotting code.
2. **V=5 ADM validation memo**: grid refinement, substrate subtraction, stage/region ledger, and packet norm status.
3. **Catch/rematch control-law memo**: compact ansatz, residual coverage, sensitivity to envelope width, and cap behavior.
4. **Source-family screen**: scalar/effective/jacket/catch components scored against ADM increments, with local amplitude and gradient diagnostics.
5. **V=10 stress appendix**: same pipeline at the edge case, reported as headroom/stress behavior rather than the primary build target.

10 Conclusion

The project is now past the first ADM translation gate. At the primary target $V = 5$, the exact-builder ADM scaffold is coherent, packet norm stays clean, live negative ADM energy density is absent, and the apparent live packet energy-density burden subtracts away as standing substrate curvature. The live service burden is concentrated in the expected catch/rematch radial momentum handoff. At $V = 10$, the same channel sharpens but does not produce a new packet-filling energy-density failure. The correct next move is an engineering prototype centered on the $V = 5$ ADM scaffold, with $V = 10$ retained as an edge-case stress appendix.

References

[1] M. Alcubierre, "The warp drive: hyper-fast travel within general relativity," *Classical and Quantum Gravity*, 11, L73-L77, 1994. arXiv version: https://arxiv.org/abs/gr-qc/0009013.

[2] M. J. Pfenning and L. H. Ford, "The unphysical nature of 'Warp Drive'," arXiv:gr-qc/9702026, 1997. https://arxiv.org/abs/gr-qc/9702026.

[3] C. Barcelo and M. Visser, "Scalar fields, energy conditions, and traversable wormholes," arXiv:gr-qc/0003025, 2000. https://arxiv.org/abs/gr-qc/0003025.

[4] A. Bobrick and G. Martire, "Introducing Physical Warp Drives," arXiv:2102.06824, 2021. https://arxiv.org/abs/2102.06824.

[5] R. Arnowitt, S. Deser, and C. W. Misner, "The Dynamics of General Relativity," in *Gravitation: An Introduction to Current Research*, ed. L. Witten, Wiley, 1962. arXiv reprint: https://arxiv.org/abs/gr-qc/0405109.

[6] E. Gourgoulhon, "3+1 Formalism and Bases of Numerical Relativity," arXiv:gr-qc/0703035, 2007. https://arxiv.org/abs/gr-qc/0703035.